

Variable Depth $H(x, y)$: Absolute Vorticity $\rightarrow$ Potential Vorticity

Rossby Palooza — Alluvial Rivers & Jet streams

Derivation companion to `numerical/dedalus_meander2/`

The one assumption we drop

The constant-depth model (`dedalus_meander`) assumed a **flat rigid bed**, $H = \text{const.}$. Dropping it (Station 1 assumption (iv)) changes the *conserved quantity*:

$$\text{flat bed: } \frac{D\zeta}{Dt} = \mathcal{C} \quad \longrightarrow \quad \text{variable } H : \quad \frac{D}{Dt} \left(\frac{\zeta}{H} \right) = \frac{1}{H} \text{curl } \mathcal{F}$$

Absolute-vorticity conservation becomes **potential-vorticity (PV) conservation**. The Rossby restoring force gains a **topographic-shear** β . Everything else – jet, erodible banks, low-Froude rigid lid – is kept.

Station 1 revisited: the mass-transport streamfunction

Low-Froude rigid lid, but now $h \approx H(x, y)$ is *not* constant. Continuity

$$\partial_t h + \nabla \cdot (h\mathbf{U}) = 0 \xrightarrow{\text{Fr}^2 \ll 1} \nabla \cdot (H\mathbf{U}) \approx 0$$

So it is the **mass flux** $H\mathbf{U}$ that is non-divergent, not $\mathbf{U}$. Introduce the **mass-transport streamfunction** Ψ' :

$$u = -\frac{1}{H} \partial_y \Psi', \quad v = +\frac{1}{H} \partial_x \Psi', \quad \zeta' = \partial_x v - \partial_y u = \nabla \cdot \left(\frac{1}{H} \nabla \Psi' \right)$$

The relative vorticity is now a **variable-coefficient elliptic** operator on Ψ' (flat bed: $\zeta' = \nabla^2 \Psi'$).

The curl: the PV-tendency equation

Curl of the shallow-water momentum equation (the surface term $-g\nabla\xi$ is a gradient, killed exactly). Using $\nabla \cdot \mathbf{U} = -\frac{1}{H}\mathbf{U} \cdot \nabla H$:

$$\partial_t \zeta' + \mathbf{U} \cdot \nabla \zeta' - \frac{\zeta'}{H} \mathbf{U} \cdot \nabla H = \text{curl } \mathcal{F} \quad \Longleftrightarrow \quad \frac{D}{Dt} \left(\frac{\zeta'}{H} \right) = \frac{1}{H} \text{curl } \mathcal{F}$$

The extra **vortex-stretching term** $-\frac{\zeta'}{H} \mathbf{U} \cdot \nabla H$ is what the flat bed did not have. It combines with base advection into a modified β – next slide.

Linearize (base bed $\bar{H}(y)$): the SAME skeleton

Base jet $\bar{u}(y)$ over base depth $\bar{H}(y)$; $\bar{\zeta} = -\bar{u}_y$. The stretching term folds into the advection, and the linear equation is **term-for-term the flat-bed model** with two replacements:

$$\partial_t \zeta' + \bar{u}(y) \partial_x \zeta' + \beta_{\text{top}}(y) \partial_x \Psi' = \text{curl } \mathcal{F}', \quad \zeta' = \nabla \cdot \left(\frac{1}{\bar{H}} \nabla \Psi' \right)$$

$$\beta_{\text{top}}(y) = \partial_y \left(\frac{\bar{\zeta}}{\bar{H}} \right) = \frac{2D}{\bar{H}} + \frac{\bar{u}_y \bar{H}_y}{\bar{H}^2} \quad (\text{flat bed } \bar{H} = 1 : \beta_{\text{top}} \rightarrow 2D, \zeta' \rightarrow \nabla^2 \Psi')$$

$\beta \rightarrow \beta_{\text{top}}$ (topographic-shear) and $\nabla^2 \rightarrow$ the $\bar{H}$ -weighted elliptic operator. Flat bed **recovers dedalus_meander exactly**.

An honest word on “topographic β ”

In a **non-rotating** river ($f = 0$) the base PV is $\bar{\zeta}/\bar{H}$ – relative vorticity only. Hence

$$D = 0 \Rightarrow \bar{\zeta} = 2Dy = 0 \Rightarrow \beta_{\text{top}} = 0 \quad \text{(a bed bump alone makes NO } \beta \text{)}.$$

The bed *modulates the shear- β* ; it does not create a standalone topographic β without planetary vorticity. A formal knob f_0 (default 0) restores true **topographic Rossby waves**:

$$\beta_{\text{top}} = \partial_y \left(\frac{\bar{\zeta} + f_0}{\bar{H}} \right) = \underbrace{\partial_y(\bar{\zeta}/\bar{H})}_{\text{shear}} - \underbrace{f_0 \bar{H}_y / \bar{H}^2}_{\text{topographic}} \quad (f_0 \neq 0 : \text{ waves even at } D = 0).$$

A survey of the group's Dedalus code found **no variable-coefficient elliptic-LHS precedent**; the one $\text{NCC} \times \text{unknown}$ on the LHS destabilised the IMEX stepper, and the docs warn that rational $1/H$ NCCs make dense matrices. Safe, preceded route (as in the QG/shallow-water scripts):

- carry ζ' as an **auxiliary field** (first-order reduction);
- multiply the elliptic constraint through by $\bar{H}^2$ to clear the $1/H$:

$$\bar{H} \nabla^2 \psi' - \bar{H}_y \partial_y \psi' - \bar{H}^2 \zeta' = 0 \quad (\bar{H} \text{ polynomial} \Rightarrow \text{all-polynomial, well-banded});$$

- multiply the PV tendency through by $\bar{H}^2$ (eigenvalue-preserving row scaling).

The “water outside the channel”: which 2nd-order effect?

Drawing the linear field on the *fixed* box $[-1, 1]$ while the banks displace by $\eta = \psi_b / U_0$ ($U_0 < 1 \Rightarrow \eta > \psi'$) leaves an $O(\text{ampl})$ -wide strip (“water not filling / water outside”). It is **purely geometric**, *not* a flow. Four second-order things, routinely conflated:

- **Geometric** $\delta_0 = ke$ (intrinsic $\rightarrow$ Cartesian): puts the field IN the meandering channel. **THIS is the gap**; `warp_fill` is its leading order. Parker 1982 keeps $O(\delta_0^3)$.
- **Nonlinear** $J(\psi', \zeta') = O(\text{ampl}^2)$: flow self-advection (fattening/skewing of the bend). Dropped – we are linear.
- **Ikeda** $\nu = b/r_0$, **kept to** $O(\nu^2)$: the FLOW’s response to bend *curvature* (superelevation $\propto F^2$, near-bank velocity), taken at $\delta_0 \rightarrow 0$. A *different parameter* – curvature, not amplitude.
- **Secondary flow** $A \approx 2.89$: 3-D helical $\partial u / \partial z$ that scours the outer bank. Depth-averaged away in both models; Ikeda re-imports it as $\eta' / H = -A \mathcal{C}' \tilde{n}$. We omit it (a bed/cross-section effect).

The gap is filled by the coordinate map (warp), not by any flow. Ikeda’s $O(\nu^2)$ is curvature-flow at zero amplitude; secondary flow is 3-D bed-scour – both are *different* things, orthogonal to this planform gap.

Along-channel bed: flow over a wavy bed

A bed varying *along* the channel, $H(x, y) = \bar{H}(y)[1 + a_x \cos(k_b x)]$ (Ikeda's alternate bars), breaks x-homogeneity – Fourier modes couple, so it is **IVP-only** (no normal modes). The base flow is fixed by **discharge conservation** $\partial_x(Hu) = 0$:

$$Hu_0 = \bar{H}(y)\bar{u}(y) \Rightarrow u_0(x, y) = \frac{\bar{H}(y)\bar{u}(y)}{H(x, y)} \quad (a_x=0 : u_0 = \bar{u}, \text{ the } \bar{H}(y) \text{ case}).$$

Flow **speeds over bars, slows over pools**. With prognostic PV $q' = \zeta'/H$ the system stays polynomial-coefficient:

$$H\partial_t q' + \bar{H}\bar{u}\partial_x q' - \partial_y \Psi' \partial_x \bar{q}_0 + \partial_x \Psi' \partial_y \bar{q}_0 + \gamma q' = 0, \quad H\nabla^2 \Psi' - \partial_x H \partial_x \Psi' - \partial_y H \partial_y \Psi' - H^3 q' = 0.$$

Bed↔bank resonance (default $k_b = k$, bar wavelength = meander wavelength = the Ikeda bar↔bend coupling): at $a_x=0.25$ the bed shifts σ $0.10 \rightarrow 0.11$, c $-0.26 \rightarrow -0.19$ (enhanced growth, slower upstream march).

Verification ladder (all green)

check	what	result
flat-bed reduction	variable- H GEP vs VL.channel_modes ($H=1$)	0.0 (identical)
$d3$ EVP vs GEP	flat + bumped bed, both closures	4.4×10^{-9}
topographic Rossby	$f_0=0$: $\beta_{\text{top}}=0$; $f_0 \neq 0$: propagating wave	as predicted
IVP vs EVP	measured σ, c , flat + bumped bed	0.1%
along-channel bed	build_ivp_Hxy at $a_x=0$ vs $\bar{H}(y)$ EVP	0.1%

Two independent discretisations (Dedalus Chebyshev spectral & an FD+Richardson GEP) agree to $\sim 10^{-9}$ – the $\bar{H}^2$ -weighted PV reformulation is correct.

- A deeper **thalweg** $H(y) = 1 + a_H(1 - y^2)$ shifts the growth band and the **group-velocity flip** k_g (the upstream/downstream momentum boundary) to the right – modestly, because a *symmetric* bump partly cancels in β_{top} .
- **Along-channel bedforms** $H(x, y)$ **done** (discharge-conserving base flow): at bed $\leftrightarrow$ bank resonance the bars measurably force the bends.
- **Nonlinear** $J(\psi', \zeta')$ **done** (--nonlinear): small ampl reduces to linear (self-test Stage 6); finite ampl generates the **3rd-harmonic fattening/skewing** $\propto \text{ampl}^2$ (Parker 1982). Deep saturation needs ∇^8 hyperviscosity (as in shutts_1983); the weakly-nonlinear window is clean.
- *Next*: secondary flow $A \approx 2.89$ = curvature-induced bed $\eta' = -A\mathcal{C}'\tilde{n}$ (Ikeda's mechanism) coupling meander curvature $\mathcal{C}' \propto \partial_x^2 \psi_b$ to the bed – a curvature $\rightarrow$ bed feedback grafted onto the channel-Rossby model.
- Core is one Dedalus driver (meander_driver.py); all figures/movies in postprocessing/. Flat bed always recovers dedalus_meander.